

Part 1

<https://studio.edgeimpulse.com/public/996852/live>

1. Raw inputs: accX, accY, accZ, 3 channel inputs are used
2. NN classifier features:
 - a. Spectral features (specifically statistical and frequency-domain metrics derived from time-series accelerometer data).
 - b. There are 33 input features in total.
 - c. accX RMS, accX Peak 1 Height, accY Peak 3 Height, accX Peak 1 Freq, accY RMS
3. NN classifier outputs:
 - a. Overall accuracy is 100% with a loss of 0.01

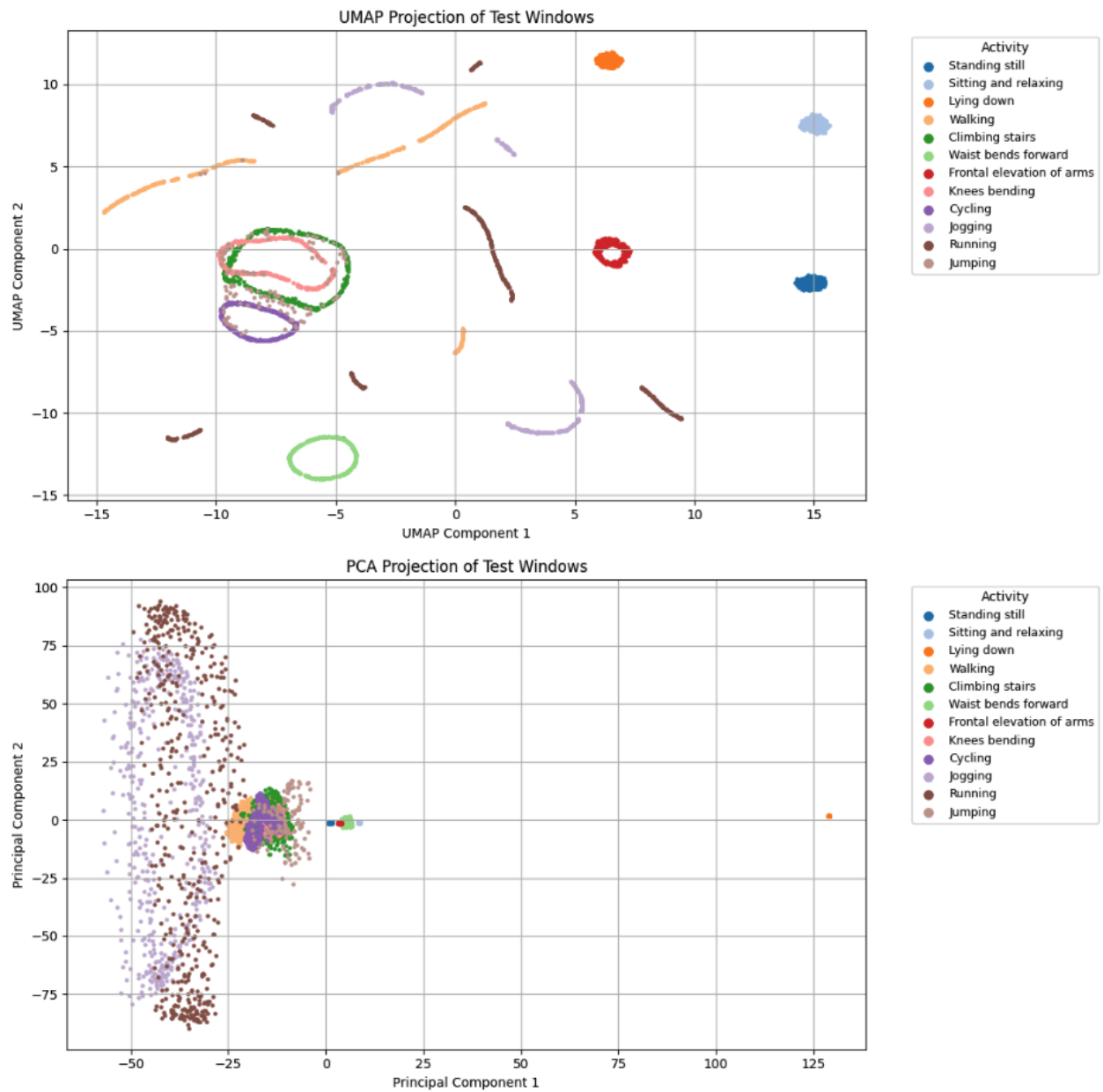
	NOMINAL	OFF
NOMINAL	100%	0%
OFF	0%	100%
F1 SCORE	1.00	1.00

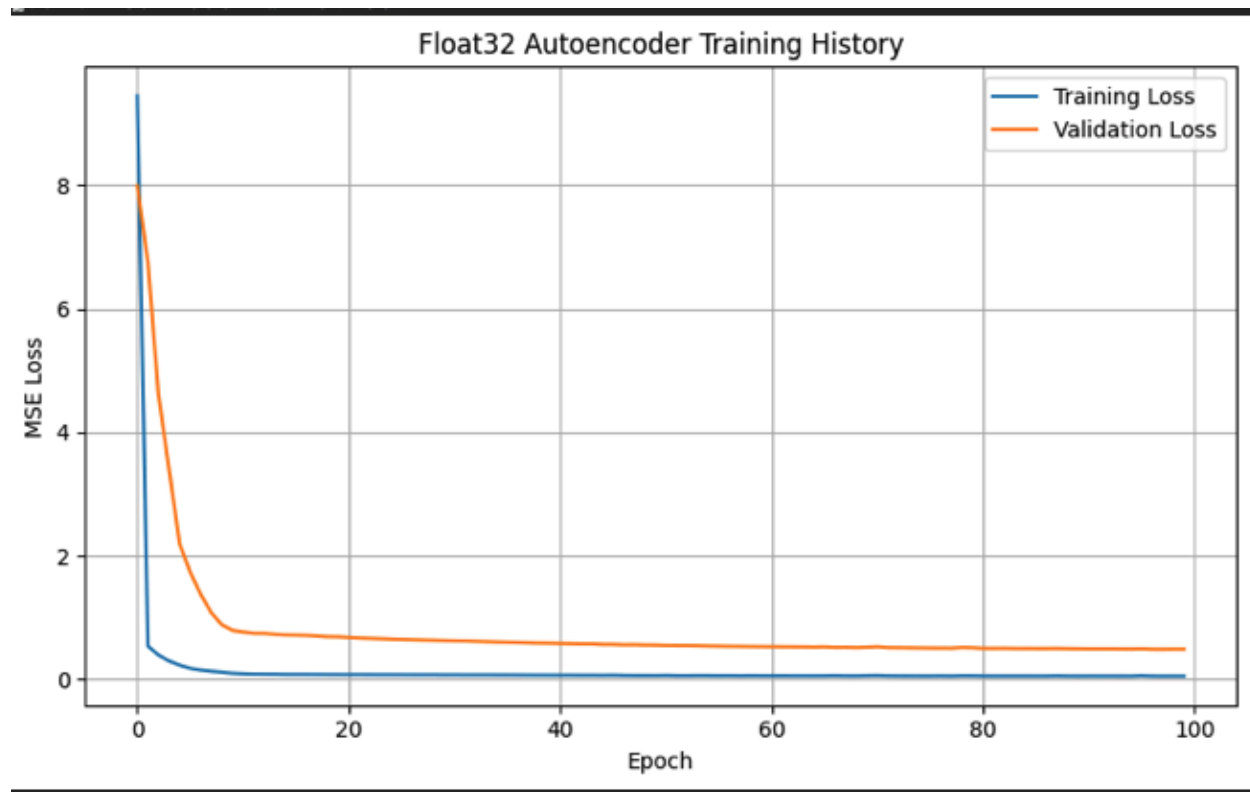
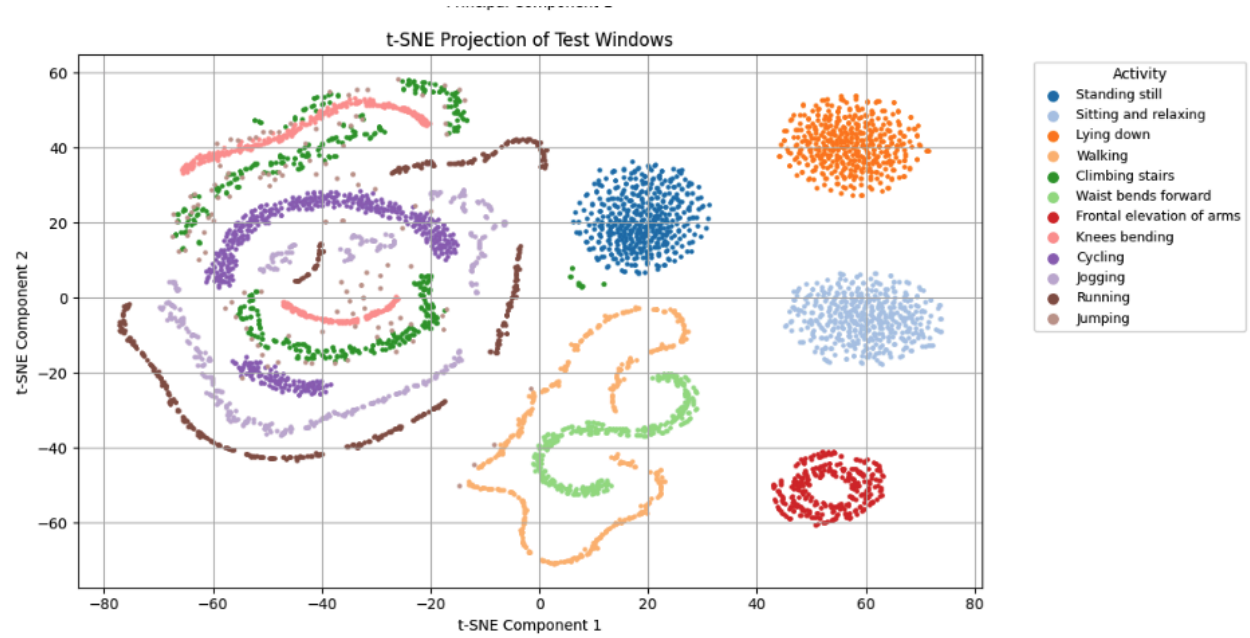
- b.
4. Anomaly Detector: We use spectral features like accX RMS, those are metrics derived from the time-series accelerometer data.
 - a. 4 features are used: accX RMS
 - b. accX Peak 1 Freq
 - c. accX Peak 1 Height
 - d. accY Peak 3 Height

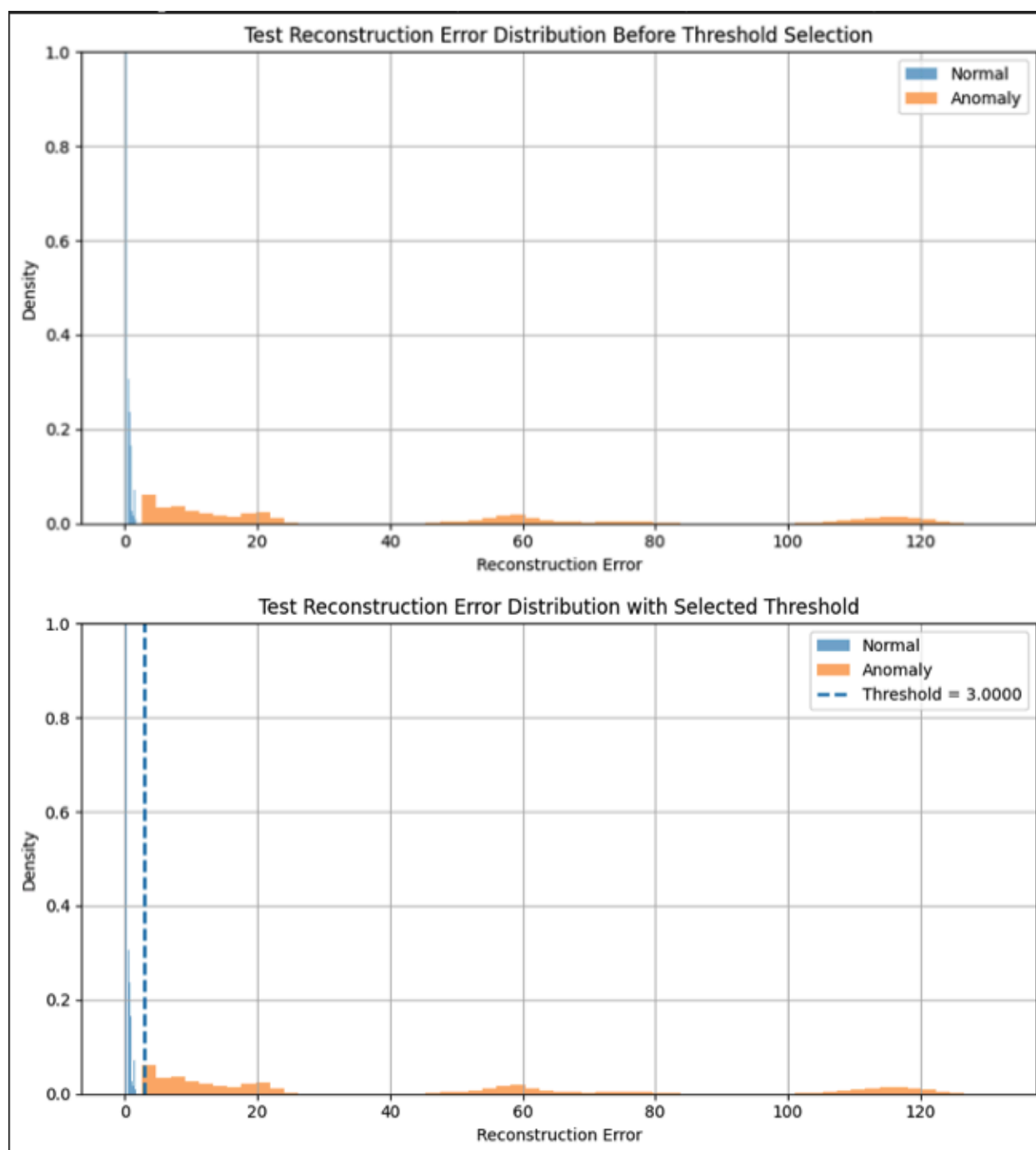
```
Anomaly prediction: 14.825054
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 6 ms, anomaly 536 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.003906
  off: 0.996094
Anomaly prediction: 21.036180
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 40 ms, inference 586 us, anomaly 6 ms, postprocessing 49 us
#Classification predictions:
  nominal: 0.898437
  off: 0.101563
Anomaly prediction: 1.103847
Starting inferencing in 2 seconds...
Sampling...
Timing: DSP 44 ms, inference 613 us, anomaly 525 us, postprocessing 49 us
#Classification predictions:
  nominal: 0.437500
  off: 0.562500
Anomaly prediction: 0.171239
Starting inferencing in 2 seconds...
```

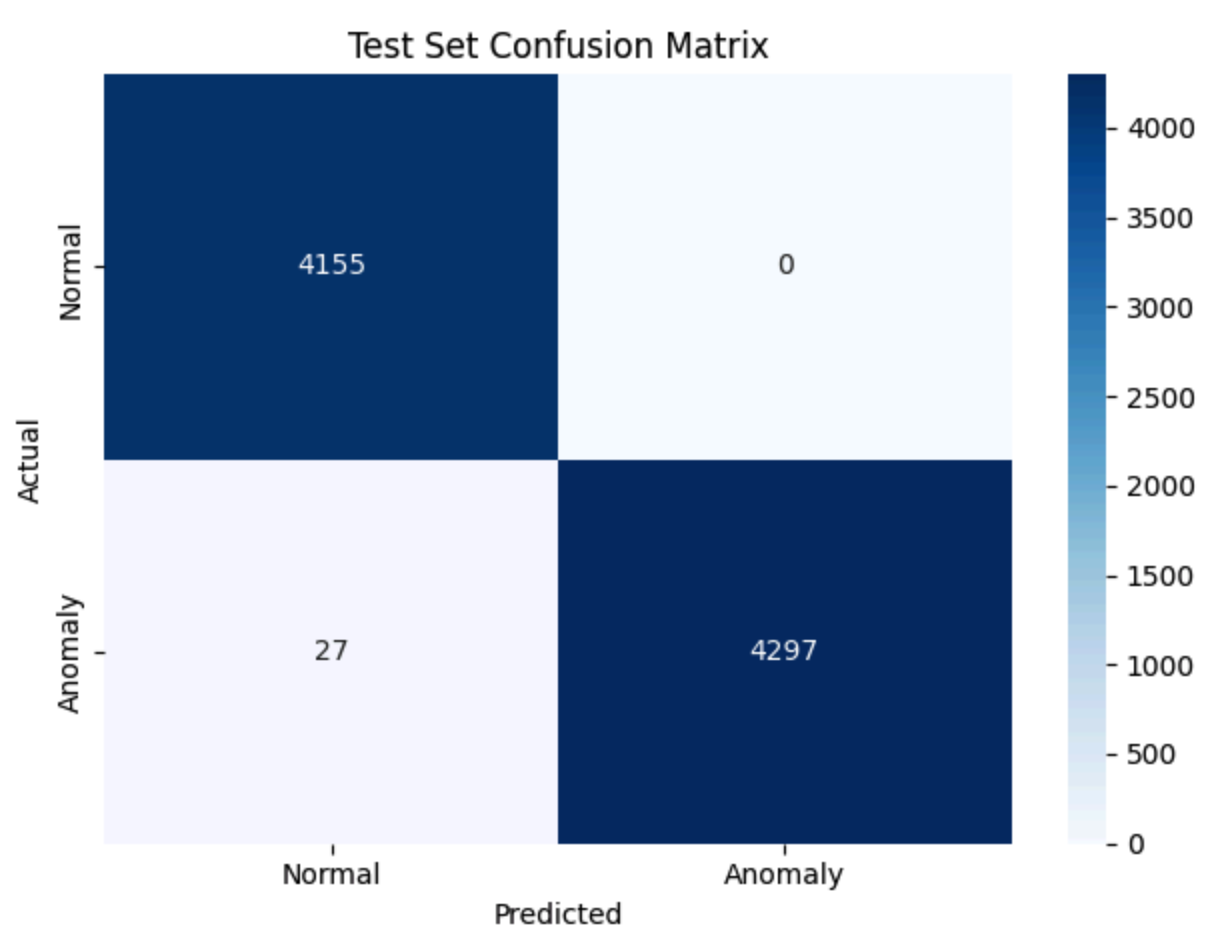
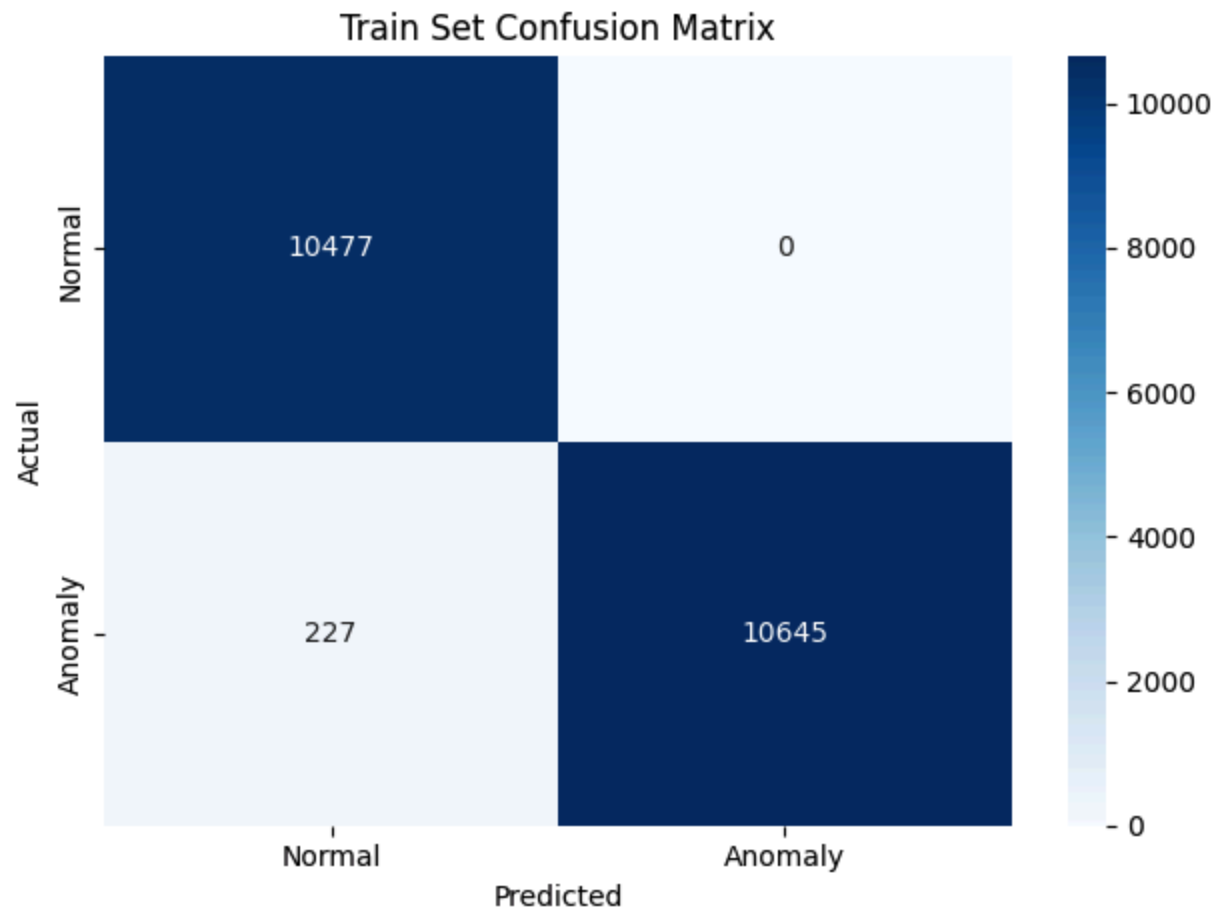
5. 
6. Inference Loop: The board executes a continuous, real-time loop. Microcontroller calculates DSP features for each window, classifies states ("nominal" or "off") via NN, and uses spatial distance metrics for the Anomaly Detector.

Part2.









Train set evaluation				
	precision	recall	f1-score	support
Normal (0)	0.98	1.00	0.99	10477
Anomaly (1)	1.00	0.98	0.99	10872
accuracy			0.99	21349
macro avg	0.99	0.99	0.99	21349
weighted avg	0.99	0.99	0.99	21349
Test set evaluation				
	precision	recall	f1-score	support
Normal (0)	0.99	1.00	1.00	4155
Anomaly (1)	1.00	0.99	1.00	4324
accuracy			1.00	8479
macro avg	1.00	1.00	1.00	8479
weighted avg	1.00	1.00	1.00	8479

The selected threshold is threshold = 3

Part3:

- What threshold did you choose, and why?
 - By observing the distribution of the reconstruction error, the threshold was chosen to be 3 which clearly distinguishes the anomaly and normal data
- How does reconstruction error help distinguish normal and anomalous activity?
 - The autoencoder is trained only on normal data, so it learns to reconstruct normal patterns well. Anomalous inputs look unfamiliar, so reconstruction is poor and error is high.
- What information from the notebook must be preserved for Arduino deployment?
 - Model weights
 - Input feature scaling parameters
 - Window size
 - Reconstruction error threshold
- Why must the Arduino sketch use the same preprocessing as the notebook?
 - The model only produces valid outputs for inputs that look like its training data. If scaling or windowing differs, the inputs are out-of-distribution and reconstruction errors become meaningless.
- What are the main limitations on a microcontroller?
 - Limited RAM
 - Integer quantization introduces small accuracy loss compared to the original float model
 - No retraining means the model can't learn new normal patterns